

1. Import necessary libraries

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns
```

Settings

```
sns.set(style="whitegrid")

plt.rcParams["figure.figsize"] = (10, 6)
```

2. Load the dataset

```
df = pd.read_csv('Titanic.csv')
```

3. Basic Info

```
print("=== Info ===") print(df.info())

print("\n=== Describe ===") print(df.describe(include='all'))

print("\n=== Missing Values ===") print(df.isnull().sum())
```

4. Value counts for categorical variables

5. Visualizations

```
print("\n=== Value Counts ===")

for col in ['gender', 'embarked', 'Pclass']: # Iterate through the desired columns
```

```
if col in df.columns: # Check if the column exists
    print(f"{col}:\n{df[col].value_counts()}\n")
else:
    print(f"Column '{col}' not found in the DataFrame.\n")
```

Histogram for Age, Fare

```
df['Age'].hist(bins=30, edgecolor='black')

plt.title('Age Distribution') plt.xlabel('Age')

plt.ylabel('Count')

plt.show()

df['Fare'].hist(bins=40, edgecolor='black')

plt.title('Fare Distribution')

plt.xlabel('Fare') plt.ylabel('Count')

plt.show()
```

Boxplot of Age by Survived

```
sns.boxplot(x='Survived', y='Age', data=df)

plt.title('Age vs Survived')

plt.show()
```

Countplot for Survived

```
sns.countplot(x='Survived', data=df)

plt.title('Survival Count')

plt.show()
```

Countplot for Pclass

```
sns.countplot(x='Pclass', hue='Survived', data=df)

plt.title('Passenger Class vs Survival')

plt.show()
```

Scatter Plot: Age vs Fare colored by Survival

```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)

plt.title('Age vs Fare by Survival') plt.xlabel('Age')

plt.ylabel('Fare')

plt.show()
```

Heatmap of correlations

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')

plt.title('Correlation Heatmap')

plt.show()
```

Pairplot (for selected numeric features)

```
sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']], hue='Survived')

plt.suptitle('Pairplot of Key Features', y=1.02)

plt.show()
```

6. Observations

Observations

1. Age is right-skewed and ranges mostly between 20 and 40.
2. Most passengers paid a low fare, but some paid significantly higher.
3. Boxplot shows younger passengers had slightly higher survival chances.
4. More people from Pclass 1 survived compared to Pclass 3.
5. Heatmap shows Fare and Pclass are moderately correlated.
6. Survival is correlated with Pclass and Fare.
7. Scatter plot shows that high-fare passengers had better survival rates, often in younger age brackets. ""

7. Summary of Findings

Summary of Findings

- Survival rate was higher among females, younger passengers, and first-class travelers.
- Pclass and Fare are key indicators related to survival.
- Missing values exist in Age, Cabin, and Embarked columns.
- Feature engineering can be done using titles from names, family size, etc. ""